Day 06 | Thursday

November 5, 2026

06

Equivalent fractions and comparing lengths

9:00 AM–10:15 AM • Math • 75 min

Mixed numbers and fraction calculator entry

10:30 AM–12:00 PM • Math • 90 min

Week 2 • 2 scheduled blocks • 165 teaching minutes

Estimate. Show the method. Label units. Check the result.

Today’s exact schedule

9:00 AM–10:15 AM • 75 min

Equivalent fractions and comparing lengths

9:00 AM–9:05 AM	**Start**	5
9:05 AM–9:10 AM	**Recall**	5
9:10 AM–9:25 AM	**Teach**	15
9:25 AM–9:40 AM	**Together**	15
9:40 AM–10:00 AM	**Practise**	20
10:00 AM–10:10 AM	**Check**	10
10:10 AM–10:15 AM	**Wrap up**	5

Math • Math Week 2, Lesson 7

10:30 AM–12:00 PM • 90 min

Mixed numbers and fraction calculator entry

10:30 AM–10:35 AM	**Start**	5
10:35 AM–10:40 AM	**Recall**	5
10:40 AM–10:55 AM	**Teach**	15
10:55 AM–11:10 AM	**Together**	15
11:10 AM–11:45 AM	**Practise**	35
11:45 AM–11:55 AM	**Check**	10
11:55 AM–12:00 PM	**Wrap up**	5

Math • Math Week 2, Lesson 8

Break: 10:15 AM–10:30 AM • 15 minutes • excluded from teaching time

Equivalent fractions and comparing lengths

Today’s goal: Use equivalent fractions to order dimensions and explain the difference between two lengths.

Name one use for today’s skill.

Textbook Chapter 4 pp.59–61

Exercise Book 4A Q6–11 (p.26)

Keep your notebook, textbook, exercise book and chosen calculator ready.

Textbook Chapter 4 pp.59–61

Recall and reconnect

Show 1/2 and 8/16 on a rule. Explain why they refer to the same point.

Try first without displayed answers.

Resources: Exercise Book 4A Q6–11 (p.26)

Textbook Chapter 4 pp.59–61

Keep equivalent fractions equal

1. Multiply or divide numerator and denominator by the same nonzero number.

2. Changing the fraction’s name does not change its length.

3. Use a fraction strip or rule to show that equivalent values share one point.

4. Reduce the final answer when the question asks for lowest terms.

Textbook Chapter 4 pp.59–61

Compare pieces of the same size

1. Find a common denominator before comparing the numerators.
2. With equal denominators, the larger numerator names the longer length.
3. Subtract the equal-sized fractions to find how much longer.
4. Order labelled dimensions by their values, not by the display’s zoom.

Textbook Chapter 4 pp.59–61

Compare equal-sized fractional units

3/4 = 12/16
12/16 > 11/16
Difference = 12/16 − 11/16 = 1/16 in

The numerators can be compared because the pieces are now the same size. Confirm the two marks on a rule.

Together • Exercise Book 4A Q6–8.
Use fraction strips to explain the longer dimension.
Shape Canvas: compare labelled rectangles in inches, not screen lengths.

Textbook | page 59



Figure 4.7 $\frac{1}{2} = \frac{2}{4} = \frac{4}{8} = \frac{8}{16}$. The dashed line is at the half.

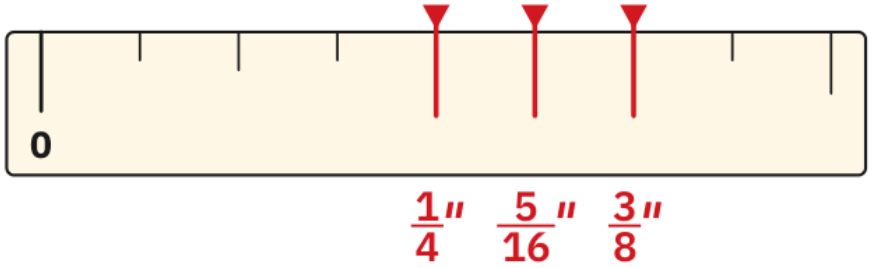
Textbook p.59 · Figure 4.7: equivalent fractions mark the same length.

Textbook Chapter 4 pp.59–61

Textbook | page 61

Example 4.5 Putting sizes in order

Three bolts have diameters of $\frac{3}{8}$ in, $\frac{5}{16}$ in and $\frac{1}{4}$ in. List them from smallest to largest.



SOLUTION

1. Change each to sixteenths: $\frac{3}{8} = \frac{6}{16}$, $\frac{5}{16}$ stays as it is, and $\frac{1}{4} = \frac{4}{16}$.
2. Put the numerators in order: 4, 5, 6.
3. The rule shows the same order from left to right.

Answer $\frac{1}{4}, \frac{5}{16}, \frac{3}{8}$

Textbook p.61 · Example 4.5: compare common denominators and confirm on the rule.

Textbook Chapter 4 pp.59–61

Spot the mistake. Explain the correction.

Is 11/16 longer than 3/4 because 11 is greater than 3?

Discuss the method before checking the answer.

Compare equal-sized parts: 3/4 = 12/16, which is longer than 11/16 by 1/16.

Textbook Chapter 4 pp.59–61

Work together

Work 4A questions 6–8 with fraction strips; confirm the longer length on a rule.

Work in pairs; alternate solver and checker.

Resources: Exercise Book 4A Q6–11 (p.26)

Textbook Chapter 4 pp.59–61

Guided practice | 4A Q6

Write $\frac{14}{32}$ and $\frac{36}{64}$ in lowest terms.

Work with a partner. Take turns explaining the next step.

Textbook Chapter 4 pp.59–61

Compare your method | 4A Q6

Write $\frac{14}{32}$ and $\frac{36}{64}$ in lowest terms.

CHECK TOGETHER

7/16 and 9/16

Textbook Chapter 4 pp.59–61

Your scheduled practice

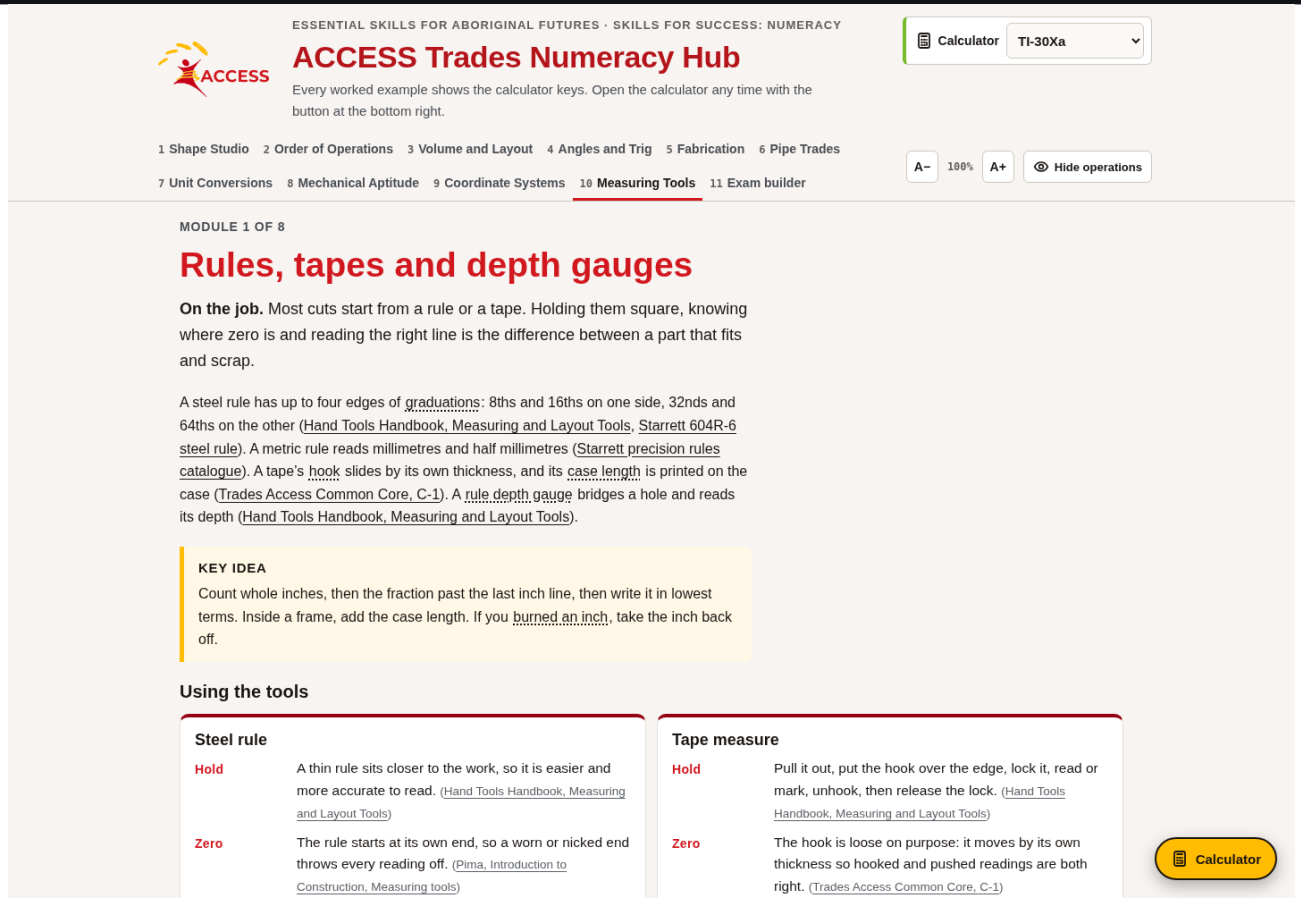
Complete 4A questions 9–11. Place three labelled dimensions in order and explain the comparison to a partner.

Show working and units; estimate, then check.

Resources: Exercise Book 4A Q6–11 (p.26)

Textbook Chapter 4 pp.59–61

Open Numeracy Hub



ESSENTIAL SKILLS FOR ABORIGINAL FUTURES · SKILLS FOR SUCCESS: NUMERACY

ACCESS Trades Numeracy Hub

Every worked example shows the calculator keys. Open the calculator any time with the button at the bottom right.

Calculator TI-30Xa

1 Shape Studio 2 Order of Operations 3 Volume and Layout 4 Angles and Trig 5 Fabrication 6 Pipe Trades 7 Unit Conversions 8 Mechanical Aptitude 9 Coordinate Systems 10 Measuring Tools 11 Exam builder

A– 100% A+ Hide operations

MODULE 1 OF 8

Rules, tapes and depth gauges

On the job. Most cuts start from a rule or a tape. Holding them square, knowing where zero is and reading the right line is the difference between a part that fits and scrap.

A steel rule has up to four edges of graduations: 8ths and 16ths on one side, 32nds and 64ths on the other (Hand Tools Handbook, Measuring and Layout Tools, Starrett 604R-6 steel rule). A metric rule reads millimetres and half millimetres (Starrett precision rules catalogue). A tape's hook slides by its own thickness, and its case length is printed on the case (Trades Access Common Core, C-1). A rule depth gauge bridges a hole and reads its depth (Hand Tools Handbook, Measuring and Layout Tools).

KEY IDEA

Count whole inches, then the fraction past the last inch line, then write it in lowest terms. Inside a frame, add the case length. If you burned an inch, take the inch back off.

Using the tools

Steel rule

Hold	A thin rule sits closer to the work, so it is easier and more accurate to read. (Hand Tools Handbook, Measuring and Layout Tools)
Zero	The rule starts at its own end, so a worn or nicked end throws every reading off. (Pima, Introduction to Construction, Measuring tools)

Tape measure

Hold	Pull it out, put the hook over the edge, lock it, read or mark, unhook, then release the lock. (Hand Tools Handbook, Measuring and Layout Tools)
Zero	The hook is loose on purpose: it moves by its own thickness so hooked and pushed readings are both right. (Trades Access Common Core, C-1)

Calculator

Navigation example. Use the values and units in the assigned task.

Textbook Chapter 4 pp.59–61

Practise in Shape Canvas

1. Supported: Use fraction strips and a rule to explain the equivalence in 4A Q6–8.

2. Independent practice: In Shape Canvas, select inches and create three labelled dimensions from Q9–11.

3. Independent checking: Order the written values using common denominators and explain them to a partner.

Textbook Chapter 4 pp.59–61

Practise from the Exercise Book | 4A Q9

Put in order from smallest: $\frac{13}{32}$, $\frac{3}{8}$, $\frac{7}{16}$.

Write the operation, working and units before checking.

Textbook Chapter 4 pp.59–61

Practise from the Exercise Book | 4A Q10

Change $\frac{37}{8}$ and $\frac{53}{16}$ to mixed numbers.

Write the operation, working and units before checking.

Textbook Chapter 4 pp.59–61

Check independently

Complete 4A questions 6 and 8, showing the common denominator.

Answer independently and show the method.

Resources: Exercise Book 4A Q6–11 (p.26)

Textbook Chapter 4 pp.59–61

Independent check | 4A Q6

Write $\frac{14}{32}$ and $\frac{36}{64}$ in lowest terms.

Answer independently. Keep worked solutions closed.

Textbook Chapter 4 pp.59–61

Independent check | 4A Q8

Which is longer, $\frac{9}{16}$ in or $\frac{5}{8}$ in? By how much?

Answer independently. Keep worked solutions closed.

Textbook Chapter 4 pp.59–61

Wrap up and choose the next step

Share one learning point and one skill to practise next.

Record a takeaway and next action.

Resources: Exercise Book 4A Q6–11 (p.26)

Textbook Chapter 4 pp.59–61

Break

10:15 AM–10:30 AM

Return ready for: Mixed numbers and fraction calculator entry

This break is excluded from teaching minutes.

Mixed numbers and fraction calculator entry

Today’s goal: Convert mixed and improper fractions and match the calculator display to the written value.

Name one use for today’s skill.

Textbook Chapter 4 pp.61–63; Chapter 2 p.34

Exercise Book 4A Q10–13 (p.26); 2A Q9 (p.14)

Keep your notebook, textbook, exercise book and chosen calculator ready.

Textbook Chapter 4 pp.61–63; Chapter 2 p.34

Recall and reconnect

Explain how many eighths are in 4 5/8. Check the meaning of numerator and denominator.

Try first without displayed answers.

Resources: Exercise Book 4A Q10–13 (p.26); 2A Q9 (p.14)

Textbook Chapter 4 pp.61–63; Chapter 2 p.34

Turn whole units into fractional units

1 A mixed number contains a whole-number part and a fraction.

2 Multiply the whole number by the denominator, then add the numerator.

3 Keep the denominator: the size of each fractional unit has not changed.

4 Write the improper fraction before checking it on the calculator.

Textbook Chapter 4 pp.61–63; Chapter 2 p.34

Turn an improper fraction back into a mixed number

1 Divide the numerator by the denominator.

2 The whole-number quotient gives the whole part.

3 The remainder becomes the numerator over the original denominator.

4 Read the calculator display aloud and compare both parts with your written value.

Textbook Chapter 4 pp.61–63; Chapter 2 p.34

Change the form—not the value

$2\ 3/8 = (2 \times 8 + 3)/8 = 19/8$

$19 \div 8 = 2$ remainder $3 \rightarrow 2\ 3/8$

The denominator stays 8 because we are still counting eighths.
Read the calculator fraction display aloud; check that whole and fractional parts match your written value.

Together • Complete 4A Q10–12.
Compare paper conversions with Numeracy Hub Calculator using your selected model and the Textbook fraction-key sequence.

Textbook | page 63

CALCULATOR Keying in $3\frac{5}{8}$

TI-30Xa

Key in $3\frac{5}{8}$ and press =

3 | a b/c | 5 | a b/c | 8 | =

Display: 3 ⌟5 ⌟8

Change it to an improper fraction

3 | a b/c | 5 | a b/c | 8 | = | 2nd | d/c

Display: 29 ⌟8

Sharp EL-531XT

Key in $3\frac{5}{8}$ and press =

3 | a b/c | 5 | a b/c | 8 | =

Display: 3 ⌜5 ⌜8

Change it to an improper fraction

3 | a b/c | 5 | a b/c | 8 | = | 2ndF | d/c

Display: 29 ⌜8

Textbook p.63 · Enter a mixed number and change it to an improper fraction.

Textbook Chapter 4 pp.61–63; Chapter 2 p.34

Textbook | page 62

Example 4.7 Mixed to improper

Change $3\frac{5}{8}$ to an improper fraction.

SOLUTION

1. Multiply the whole number by the denominator: 3 whole inches hold 3 × 8 = 24 eighths.
2. Add the numerator: 24 + 5 = 29. Write it over the same denominator.

Answer $3\frac{5}{8} = \frac{29}{8}$

Textbook p.62 · Example 4.7: whole × denominator + numerator.

Textbook Chapter 4 pp.61–63; Chapter 2 p.34

Spot the mistake. Explain the correction.

A learner changes 2 3/8 into 5/8 by adding 2 + 3. Why is that wrong?

Discuss the method before checking the answer.

Two whole units contain 16 eighths. Add 3 more eighths: (2 × 8 + 3)/8 = 19/8.

Textbook Chapter 4 pp.61–63; Chapter 2 p.34

Work together

Complete 4A questions 10–12. Compare paper conversions with calculator displays.

Work in pairs; alternate solver and checker.

Resources: Exercise Book 4A Q10–13 (p.26); 2A Q9 (p.14)

Textbook Chapter 4 pp.61–63; Chapter 2 p.34

Guided practice | 4A Q10

Change $\frac{37}{8}$ and $\frac{53}{16}$ to mixed numbers.

Work with a partner. Take turns explaining the next step.

Textbook Chapter 4 pp.61–63; Chapter 2 p.34

Compare your method | 4A Q10

Change $\frac{37}{8}$ and $\frac{53}{16}$ to mixed numbers.

CHECK TOGETHER

4 5/8 and 3 5/16

Textbook Chapter 4 pp.61–63; Chapter 2 p.34

Your scheduled practice

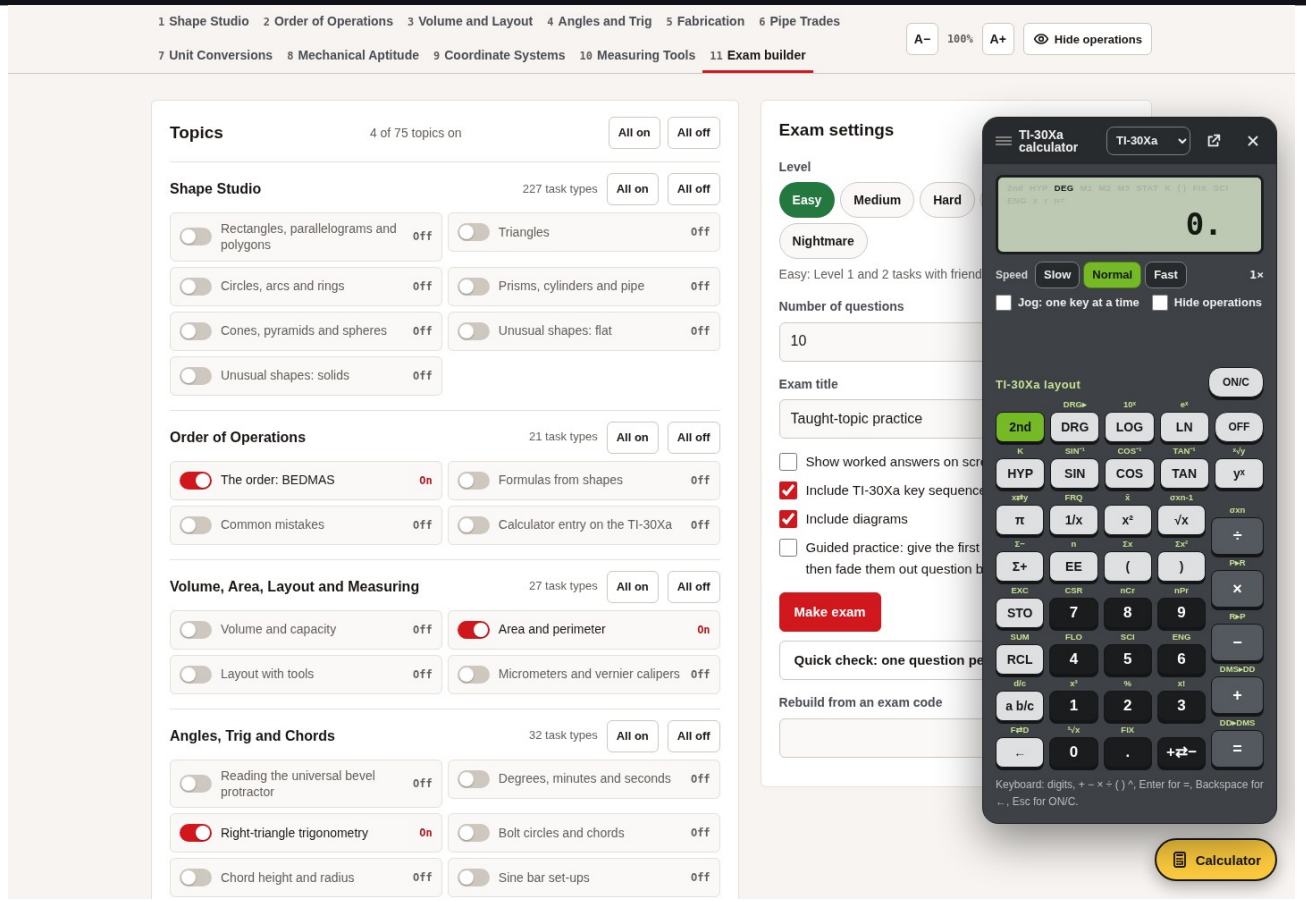
Complete question 13 and practise entering the values in questions 10–12 without changing them. Use 2A question 9 as a supported extension, not an addition-speed test.

Show working and units; estimate, then check.

Resources: Exercise Book 4A Q10–13 (p.26); 2A Q9 (p.14)

Textbook Chapter 4 pp.61–63; Chapter 2 p.34

Open Numeracy Hub



Navigation example. Use the values and units in the assigned task.

Textbook Chapter 4 pp.61–63; Chapter 2 p.34

Practise in Numeracy Hub

1. Supported: Select the learner’s Numeracy Hub calculator model and follow Textbook p.34.

2. Supported: Enter the values from 4A Q10–12 after completing each conversion on paper.

3. Independent checking: Complete Q10–11 without a calculator, then compare the display with your values.

Textbook Chapter 4 pp.61–63; Chapter 2 p.34

Practise from the Exercise Book | 4A Q13

What is the reciprocal of $2\frac{1}{2}$?

Write the operation, working and units before checking.

Textbook Chapter 4 pp.61–63; Chapter 2 p.34

Practise from the Exercise Book | 4A Q10

Change $\frac{37}{8}$ and $\frac{53}{16}$ to mixed numbers.

Write the operation, working and units before checking.

Textbook Chapter 4 pp.61–63; Chapter 2 p.34

Check independently

Complete 4A questions 10–11 without the calculator, then check.

Answer independently and show the method.

Resources: Exercise Book 4A Q10–13 (p.26); 2A Q9 (p.14)

Textbook Chapter 4 pp.61–63; Chapter 2 p.34

Independent check | 4A Q10

Change $\frac{37}{8}$ and $\frac{53}{16}$ to mixed numbers.

Answer independently. Keep worked solutions closed.

Textbook Chapter 4 pp.61–63; Chapter 2 p.34

Independent check | 4A Q11

Change $4\frac{7}{16}$ and $6\frac{3}{4}$ to improper fractions.

Answer independently. Keep worked solutions closed.

Textbook Chapter 4 pp.61–63; Chapter 2 p.34

Wrap up and choose the next step

Share one learning point and one skill to practise next.

Record a takeaway and next action.

Resources: Exercise Book 4A Q10–13 (p.26); 2A Q9 (p.14)

Textbook Chapter 4 pp.61–63; Chapter 2 p.34